Assessing Climate Change Adaptive Capacity using geospatial techniques: Application to a Coastal Upazila in Bangladesh






Muhammad Al-Amin Hoque[1,2*], Mehedi Hasan[1], Hasina Akther[1], Maliha Sanzana Misty[1]

[1] Department of Geography and Environment, Jagannath University, Dhaka-1100, Bangladesh

[2] Centre for Advanced Modelling and Geospatial Information Systems (CAMGIS), School of Civil and Environmental Engineering, University of Technology Sydney, Ultimo, NSW 2007, Australia.

জগন্নাথ বিশ্ববিদ্যালয়
Jagannath University

UNIVERSITY OF TECHNOLOGY SYDNEY

Outline

1. Introduction
2. Research Aim and Objectives
3. Study Area
4. Methods
5. Results
6. Conclusions

Introduction: Climate Change and Coastal Vulnerability

- Bangladesh's coastal region faces increasing vulnerability due to climate-induced hazards such as cyclones, sea-level rise, salinity, coastal flood and erosion.
- These hazards threaten livelihoods, infrastructure, and ecosystems.
- ~35 million people live in the coastal belt, many depending on agriculture and fisheries.
- Assessing adaptive capacity is essential for building resilience.



Introduction: Impacts of Climate Change in Coastal Bangladesh

Cyclones: Loss of lives, homes, and crops.

Sea-level rise: Coastal inundation and displacement.

Salinity intrusion: Soil degradation and drinking water scarcity.

Flooding: Damage to agriculture and infrastructure.

Erosion: Land loss and community relocation

Cyclone Sidr, Bangladesh, 2007



Coastal erosion



Coastal flooding



Salinity intrusion



Introduction: Adaptive capacity mapping

- Adaptive capacity reflects the ability of a community to anticipate, cope, and recover from such stresses.
- Mapping of climate change adaptation capacity
- A spatial multi-criteria integrated adapting capacity mapping approach at local scales (1-5 km) is essential for producing sufficiently detailed adaptative capacity map.
- The adaptive capacity map could be used for resource allocation, disaster risk reduction, and climate resilience planning.
- A geospatial multi-criteria assessment technique in the form of analytical hierarchy process (AHP) is a potentially effective approach for mapping climate change adaptive capacity.

Research Aim and Objectives

Aim:

Develops and applies a geospatial, multi-criteria framework to quantify and map climate change adaptive capacity at the local scale, using Kalapara Upazila in Patuakhali District, Bangladesh.

Objectives:

(1) Develop a spatial climate change adaptation capacity mapping approach that integrates multi-criteria for climate change adaptation capacity at the local scale covering <1000 km^2;

(2) Examine the developed approach for assessing climate change adaptation capacity at the local scale in Kalapara Upazila, Bangladesh; and

(3) Evaluate the validation of spatial climate change adaptation assessment approach.

Study Area

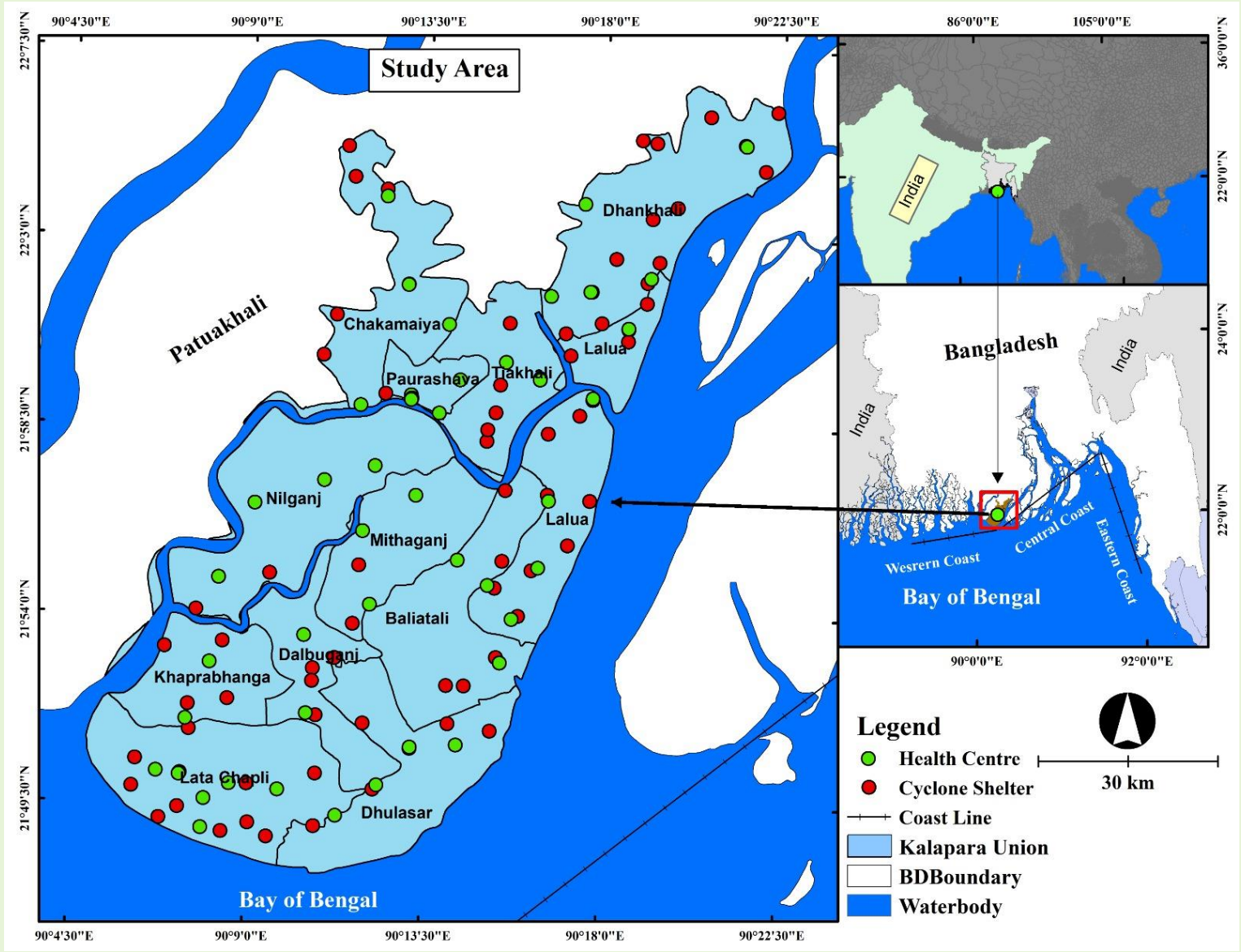
Location: Kalapara Upazila, Patuakhali District, southwestern Bangladesh.

Area: Approximately **491.9 km²**.

Population: Around **238,000 (Census 2022)**.

Features: Prone to **cyclones, tidal surges, salinity intrusion, and riverbank erosion**.

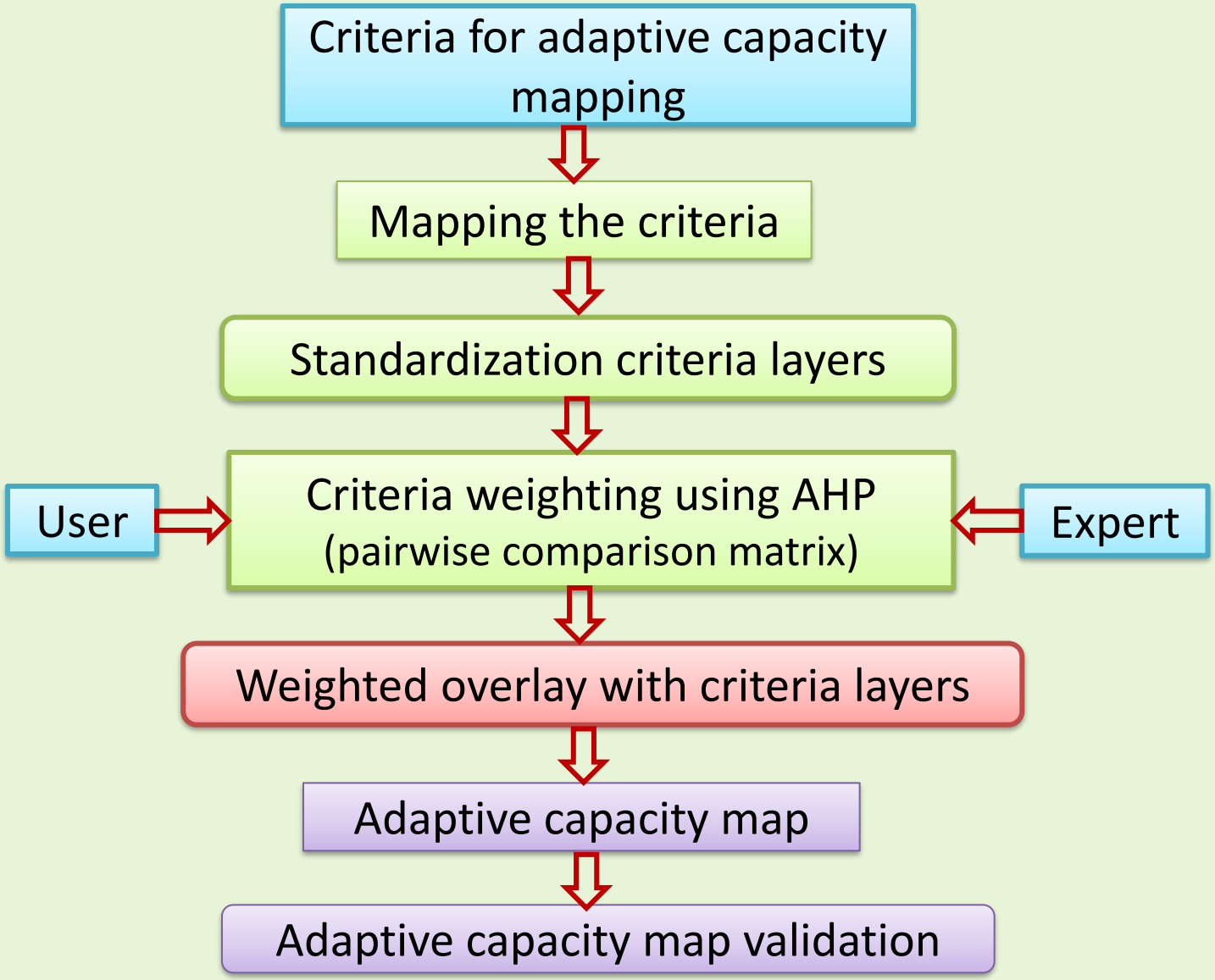
Coastal setting: Directly adjacent to the **Bay of Bengal**, including Kuakata sea beach and several small islands.



Methods: Data

Data source	Period	Mapping output
Bangladesh Bureau of Statistics (BBS)	2021–2022	Population Density, Literacy Rate, Socio-Economic Status Index (SI)
Department of Disaster Management (DDM), Local Government Engineering Department (LGED)	2023	Cyclone Shelter Density, Road Density,, Early Warning System
LGED, Institute of Water Modelling (IWM), Sentinel-2	2022–2025	Settlement Density, Land use and Land cover, Inland Pond Density
GEE (Sentinel-2), Google Earth Pro	2025	Distance to Coastal Forest, Distance to Embankments
Bangladesh Meteorological Department (BMD)	1970-2025	Storm Surge height
Directorate General of Health Services (DGHS)	2025	Health Centre density

Methods: Spatial data processing flow



Methods: Criteria selection

- Following the IPCC AR6 conceptualization of adaptive capacity and reviewing the literature
- Availability of data and their relevance and influence on climate change adaptive capacity
- 14 criteria were selected across five capital domains—**human, social, natural, physical, and financial**.

Capital Type (IPCC Framework)	Indicators / Criteria
Human Capital	Literacy Rate (LR); Population Density (PD)
Social Capital	Early Warning System (EWS) Index
Natural Capital	Storm Surge height (SSH); Distance to Coast (DC); Inland Pond Density (IPD); Land Use / Land Cover (LULC); Distance to coastal forest (DCF)
Physical Capital	Road Density (RD); Settlement Density (SD); Cyclone Shelter Density (SSD); Health Centre Density (HCD); Distance to Embankment (DE)
Financial Capital	Socio-Economic Status (SES) Index

Methods: Criteria mapping, alternatives ranking and standardization

- Spatial thematic layers of each selected criterion was generated by mapping the alternatives of each criterion using ArcGIS Pro 3.4 software

- Ranking was performed on the mapped alternatives of each spatial criterion layer providing the adaptive capacity levels (1 to 5).

- Criteria were standardized bring to values within 0 to 1 to apply AHP based weight based on the following formula.

$$P = (X - \min/\max - \min)$$

Methods: Criteria weighting using AHP

- Pairwise comparison matrix within AHP
- Criteria ratings based on relative importance scale by Saaty
- 5 experts and user judgements used for ratings

Relative importance scale by Saaty (1988, p.112)

Extremely	Very More strongly	Less Strongly	Moderately		Moderately	Very Less Strongly	More strongly	Extremely
Less important				Equally Important	More important			
1/9	1/7	1/5	1/3	1	3	5	7	9

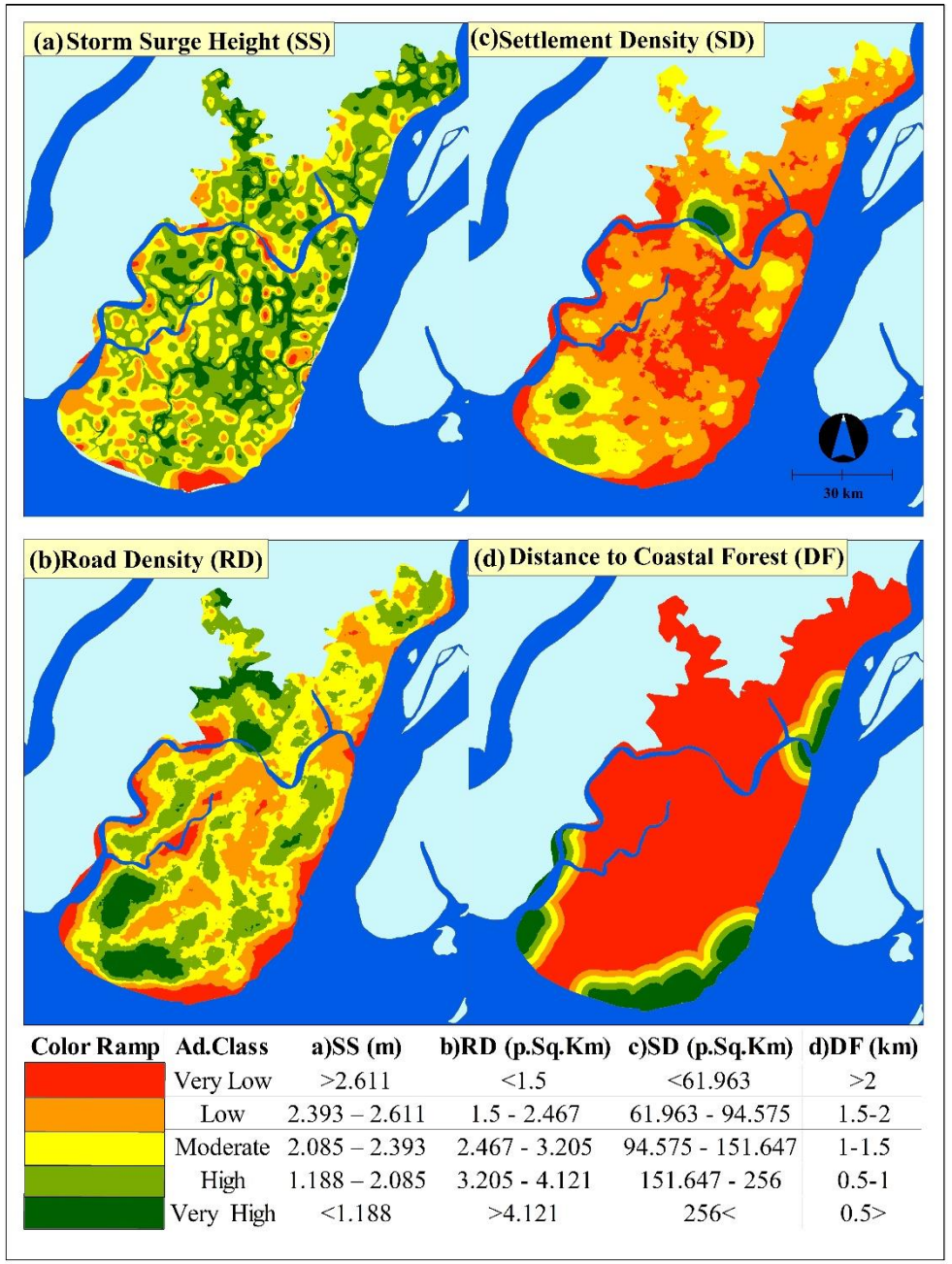
Methods: Adaptive capacity mapping and validation

- Weighted overlay with criteria layers

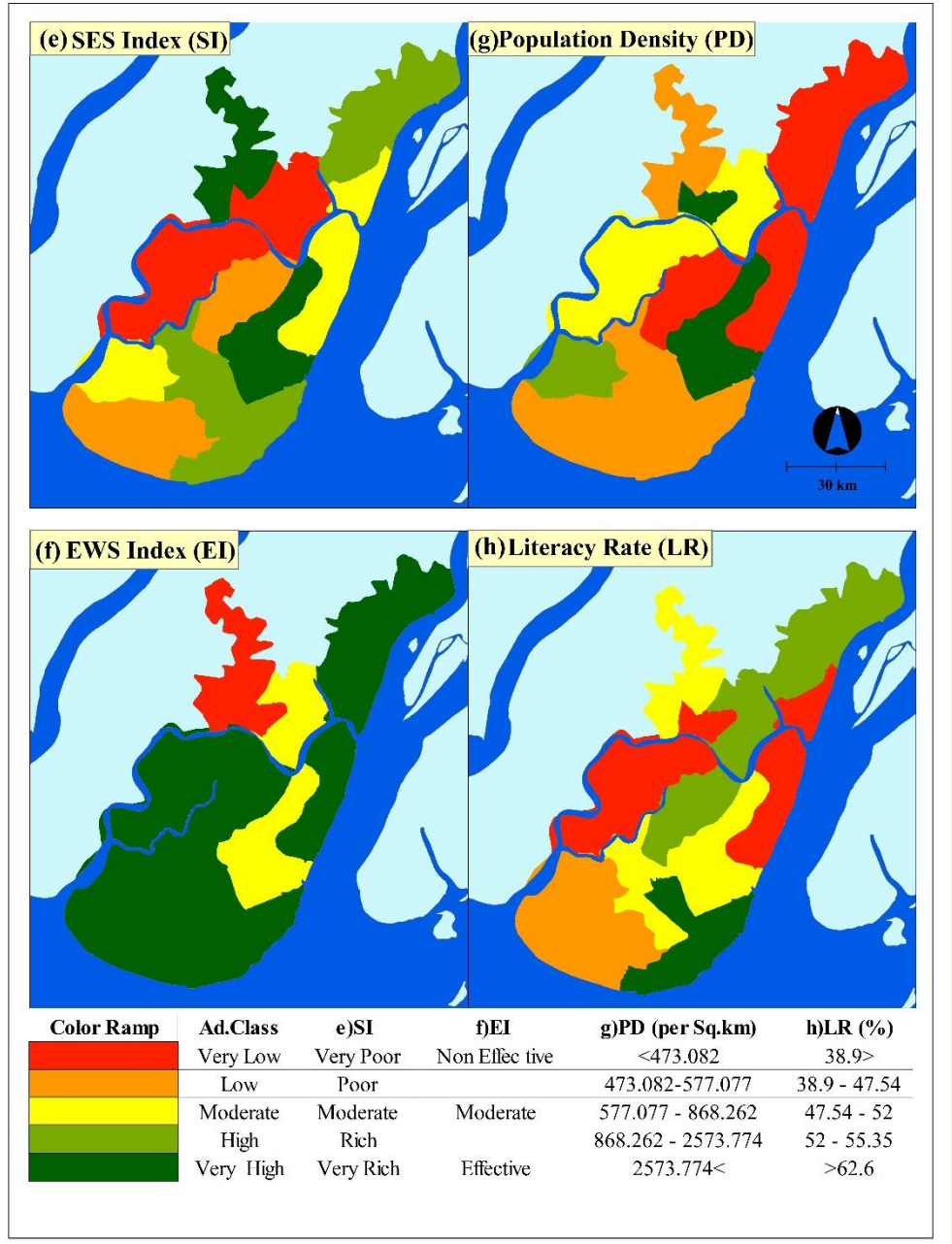
- Adaptive capacity index values were categorised into five classes classes (i.e. very low, low, moderate, high and very high).

- Validation of adaptation capacity map has been conducted using ROC and AUC curve using the validation points (about 150) collected from the field

Results: Criteria mapping

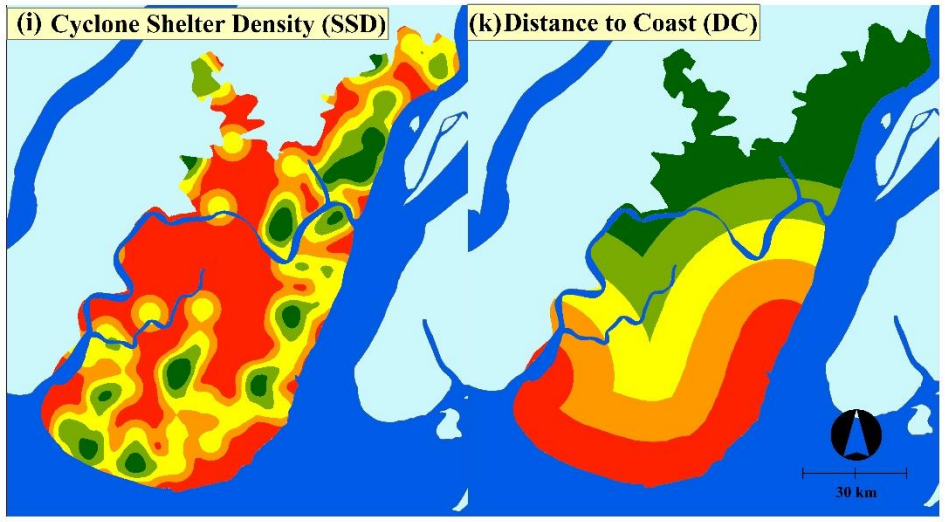


Color Ramp	Ad.Class	a)SS (m)	b)RD (p.Sq.Km)	c)SD (p.Sq.Km)	d)DF (km)
	Very Low	>2.611	<1.5	<61.963	>2
	Low	2.393 – 2.611	1.5 - 2.467	61.963 - 94.575	1.5-2
	Moderate	2.085 – 2.393	2.467 - 3.205	94.575 - 151.647	1-1.5
	High	1.188 – 2.085	3.205 - 4.121	151.647 - 256	0.5-1
	Very High	<1.188	>4.121	256<	0.5>

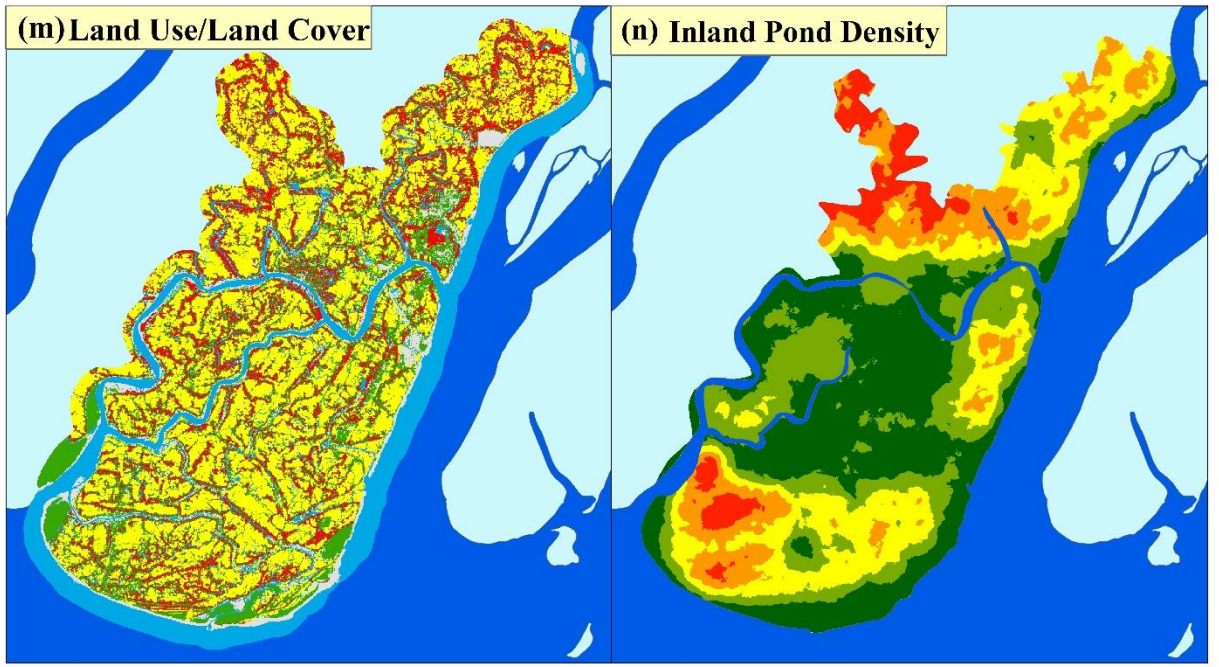


Color Ramp	Ad.Class	e)SI	f)EI	g)PD (per Sq.km)	h)LR (%)
	Very Low	Very Poor	Non Effec tive	<473.082	38.9>
	Low	Poor		473.082-577.077	38.9 - 47.54
	Moderate	Moderate	Moderate	577.077 - 868.262	47.54 - 52
	High	Rich		868.262 - 2573.774	52 - 55.35
	Very High	Very Rich	Effective	2573.774<	>62.6

Results: Criteria mapping

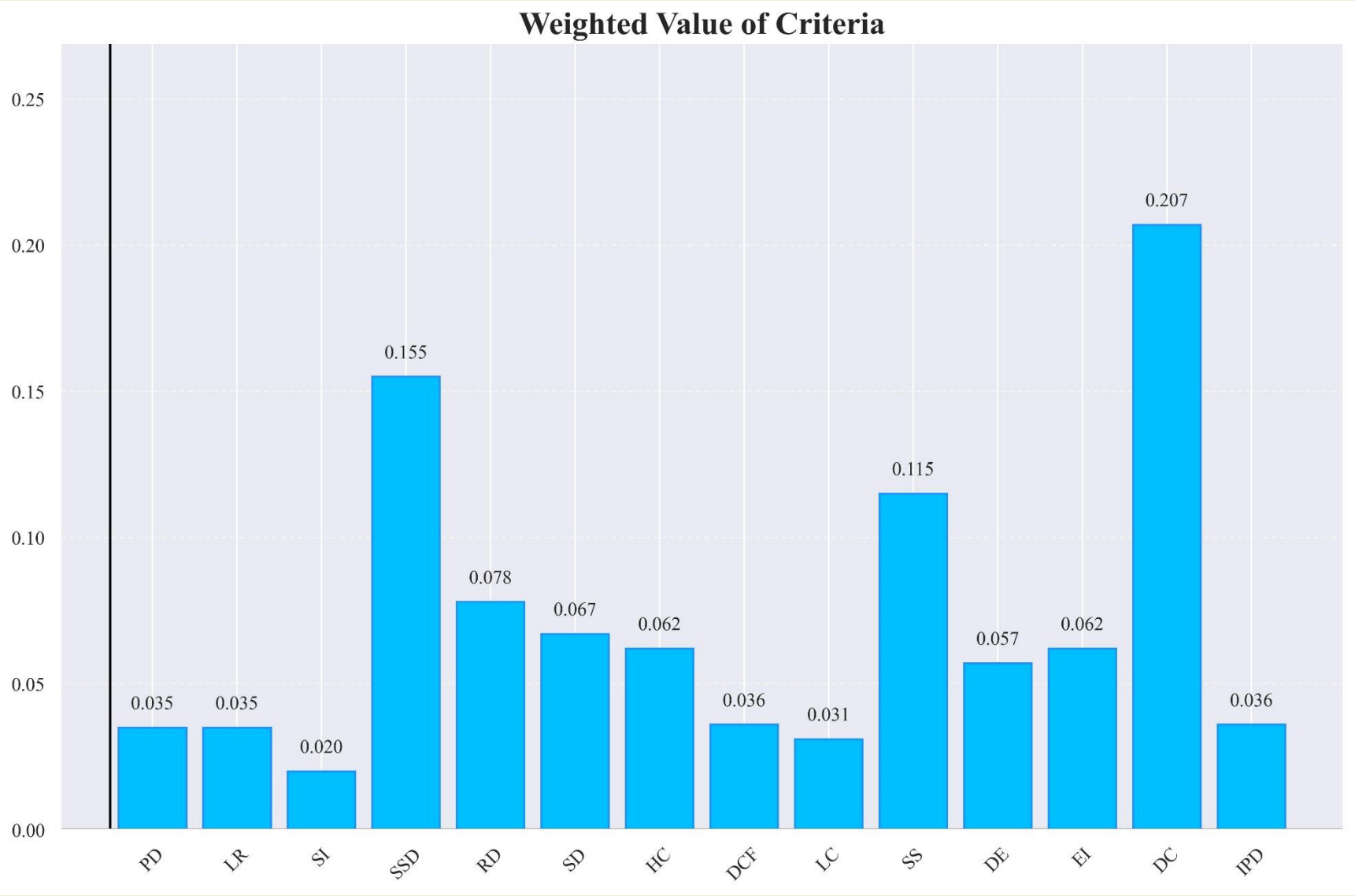


Color Ramp	Ad. Class	i)SSD (Per Sq.Km)	j)HC(km)	k)DC (Km)	l)DE (km)
	Very Low	<0.063	>2.683	<3	>2
	Low	0.063 - 0.167	1.982 - 2.683	3.0 - 6.0	1.5-2
	Moderate	0.167 - 0.273	1.401 - 1.982	6.0 - 9.0	1-1.5
	High	0.273 - 0.397	0.837 - 1,401	9.0 - 12	0.5-1
	Very High	>0.397	<0.837	12<	0.5>



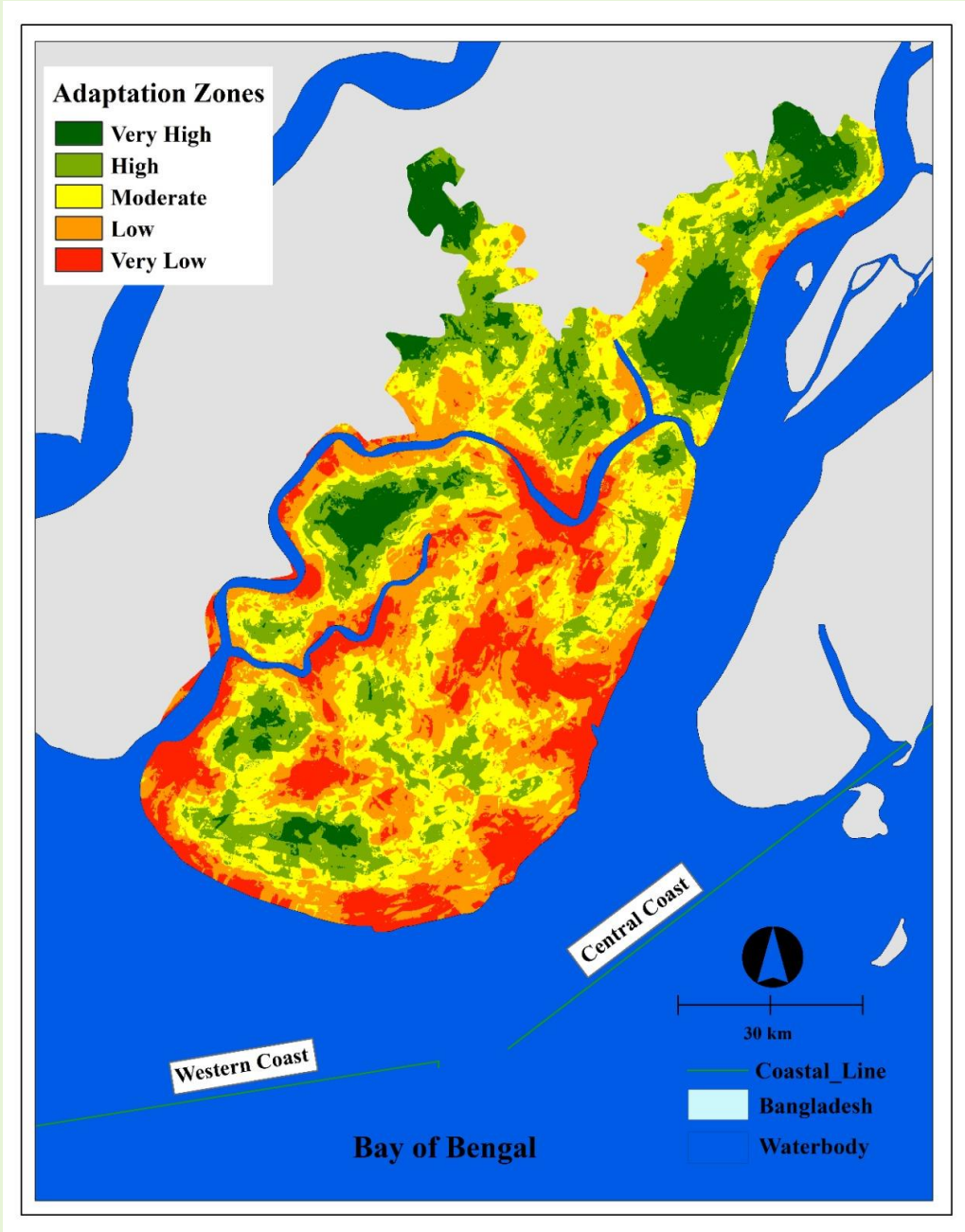
Results: Criteria weighting using AHP

- Weights of criteria derived from pairwise comparison matrix.
- The consistency ratio (CR) of comparison was 0.05 and within an acceptable range < 0.10.



Results: Adaptive capacity mapping

- 10% of area was in the very high adaptive capacity zone; toward the inland and central regions
- 49 % of the area was in the moderate to high adaptive capacity zone
- The area classified as a low and very low adaptive capacity zone was 26% and 15%, respectively; highly exposed coastal and riverine zones



Conclusions

- The adaptation capacity map provides a detailed source of spatial information that can be used used for resource allocation, disaster risk reduction, and climate resilience planning.

- The study demonstrates the capability of geospatial approach for mapping climate change adaptative capacity at the local scale.

- The validation AUC Value for the adaptation capacity mapping was 89%.

Future work will focus on:

- Apply machine learning for adaptive capacity prediction.






THANK YOU FOR YOUR ATTENTION!

Dr. Muhammad Al-Amin Hoque
Hoque@geography.jnu.ac.bd

Acknowledgements

জগন্নাথ বিশ্ববিদ্যালয়
Jagannath University

UNIVERSITY OF TECHNOLOGY SYDNEY